

Case Contents Guide -- Brachial Plexus Lecture

ANNOTATED PRESENTER GUIDE -- Why Every Finding Is What It Is

Print this. Use it to answer resident questions on the fly.

CASE 1: Severe Shoulder Pain Followed by Weakness

Dx: Parsonage-Turner Syndrome

DIFFERENTIALS THE SYSTEM ASKS ABOUT:

1. **Parsonage-Turner Syndrome** (correct)
2. **C5-C6 Radiculopathy** -- ruled out because paraspinals are normal and the pattern is patchy (not following a single root)
3. **Upper Trunk Plexopathy** -- ruled out because a trunk lesion would involve ALL muscles from that trunk. PTS affects individual nerves from different trunks/cords
4. **Rotator Cuff Tear** -- ruled out because a tear causes pain/ROM limitation but NOT neurogenic findings on EMG (no fibs, no recruitment changes)
5. **Mononeuritis Multiplex** -- ruled out because that's usually vasculitis with systemic symptoms. PTS has the viral prodrome + focal shoulder girdle predilection

WHY EACH PHYSICAL EXAM FINDING EXISTS:

Finding	Why
Deltoid 2/5	Axillary nerve (C5-C6) is one of the nerves inflamed in PTS. Deltoid is its main muscle.
Infraspinatus 2/5	Suprascapular nerve (C5-C6) is another commonly hit nerve in PTS. Infraspinatus does external rotation.
Biceps 3/5	Musculocutaneous nerve (C5-C6) is affected. Biceps = elbow flexion + supination. Less severe than deltoid because the inflammation is patchy.
Hand 5/5	The C8-T1 muscles (APB, FDI) are SPARED. PTS preferentially hits shoulder girdle nerves, not distal hand nerves. This rules out a pan-plexopathy.
Patchy numbness lateral shoulder	This is the axillary nerve sensory territory (the "regimental badge area"). Sensory loss is minor in PTS compared to motor -- it's primarily a motor condition.
Biceps reflex 1+ (diminished)	The biceps reflex arc goes through C5-C6. The musculocutaneous nerve is affected, so the efferent limb of the reflex is weakened.
Triceps reflex 2+ (normal)	C7 is spared. This helps narrow to C5-C6 territory.
Negative Spurling's	Spurling's compresses the neural foramen. Negative = no radiculopathy. PTS is a plexus/nerve-level disease, not a spine disease.

WHY EACH NCS FINDING IS WHAT IT IS:

Study	Value	Why
Median sensory 45 uV (normal)	Median nerve is NOT one of the nerves affected in this patient's PTS. C6-C7 fibers through the median are intact.	
Ulnar sensory 35 uV (normal)	Ulnar nerve (C8-T1) is not involved. Lower trunk is spared.	
Radial sensory 25 uV (normal)	Radial sensory (C6) is intact. The inflammation didn't hit this nerve.	
LABC 5 uV (ABNORMAL)	The LABC is the sensory branch of the musculocutaneous nerve. Since the musculocutaneous is inflamed (biceps is weak), its sensory branch has undergone Wallerian degeneration. This confirms the lesion is POST-GANGLIONIC -- distal to the DRG. If this were a C6 root lesion, the LABC would be normal (DRG intact).	
Axillary motor 1.5 mV (LOW)	The axillary nerve is severely affected. Low CMAP = axonal loss. The motor axons from the axillary nerve have degenerated, so fewer muscle fibers fire when you stimulate.	
Musculocutaneous motor 2.8 mV (LOW)	Same principle -- musculocutaneous nerve has axonal loss. 2.8 mV is reduced but not absent, consistent with partial injury (some axons survived).	
Median motor 12 mV (normal)	Median motor (APB, C8-T1) is fine. The median nerve trunk is not involved.	
Ulnar motor 14 mV (normal)	Ulnar motor (ADM, C8-T1) is fine. Lower trunk completely spared.	

WHY EACH EMG FINDING IS WHAT IT IS:

Muscle	Finding	Why
APB (Median, C8-T1)	Normal	C8-T1 muscles are spared. This is a DISTRACTOR -- if APB were abnormal, you'd think lower trunk or C8 root, not PTS.
FDI (Ulnar, C8-T1)	Normal	Same -- lower trunk territory intact.
EIP (Radial, C7-C8)	Normal	C7-C8 muscles via radial nerve are fine.
Deltoid (Axillary, C5-C6)	3+ fibs, reduced recruitment	Axillary nerve is inflamed. The motor axons have died, so the muscle fibers they innervated are now orphaned = fibrillations. Reduced recruitment = fewer motor units available to fire.
Infraspinatus (Suprascapular, C5-C6)	3+ fibs, reduced recruitment	Suprascapular nerve is a CLASSIC target in PTS. Same mechanism -- axonal loss causes denervation.
Serratus Anterior (Long Thoracic, C5-C7)	2+ fibs, reduced recruitment	Long thoracic nerve is THE most commonly affected nerve in PTS. This causes winged scapula clinically. Less severe fibs (2+ vs 3+) suggests milder involvement of this nerve.
Supraspinatus (Suprascapular, C5-C6)	3+ fibs, reduced recruitment	Same nerve as infraspinatus. Both suprascapular-innervated muscles are hit.
Pronator Teres (Median, C6-C7)	NORMAL	THIS IS THE PATCHY CLUE. Pronator teres is a C6-C7 muscle via the median nerve. If this were a C5-C6 root lesion, pronator teres might be affected (it shares C6). But it's NORMAL because PTS hits individual nerves, not roots. The median nerve was spared while the axillary and suprascapular were hit.
Biceps (Musculocutaneous, C5-C6)	2+ fibs, reduced recruitment	Musculocutaneous nerve is inflamed. Less severe than deltoid (2+ vs 3+) -- the patchy nature means different nerves are affected to different degrees.
C5-C6 Paraspinals	NORMAL (0 fibs)	This is CRITICAL. Paraspinals are innervated by the posterior primary ramus, which branches off the nerve root almost immediately. If the lesion were at the ROOT level, paraspinals would show fibrillations. Normal paraspinals = the lesion is DISTAL to the root = post-ganglionic = plexus or peripheral nerve level.

CASE 2: Hand Weakness After a Fall

Dx: Klumpke's Palsy (Lower Trunk Plexopathy)

DIFFERENTIALS:

1. **Lower Trunk Plexopathy (Klumpke's)** (correct)
2. **C8-T1 Radiculopathy** -- ruled out because MAC SNAP and Ulnar SNAP are abnormal (post-ganglionic). In radiculopathy (pre-ganglionic), these SNAPs would be normal.
3. **Combined Ulnar + Median Neuropathy** -- ruled out because sensory loss extends to medial arm/forearm (MAC territory) AND EIP (radial nerve) is weak. Two peripheral nerve lesions wouldn't explain MAC involvement or radial territory weakness.
4. **Thoracic Outlet Syndrome** -- ruled out because TOS is chronic/positional. This is acute traction injury.
5. **T1 Root Avulsion** -- ruled out because NO Horner's syndrome. T1 avulsion disrupts sympathetic chain = Horner's.

WHY EACH PHYSICAL EXAM FINDING:

Finding	Why
Thenar AND hypothenar atrophy	Thenar = median (C8-T1). Hypothenar = ulnar (C8-T1). BOTH atrophied = lesion crosses the median/ulnar boundary. Must be proximal to where they separate = lower trunk, medial cord, or C8-T1 roots.
Claw hand (MCP hyperextension, IP flexion digits 4-5)	Lumbricals 3-4 and interossei (ulnar nerve) are weak. Without lumbricals to flex the MCPs, the extensors pull the MCPs into hyperextension. Without interossei to extend the IPs, the FDP pulls IPs into flexion. Classic "intrinsic minus" posture.
APB 2/5, FDI 2/5, ADM 2/5	All three are C8-T1 muscles from different nerves (APB=median, FDI=ulnar, ADM=ulnar). All weak = C8-T1 level lesion.
FDP 3/5, FCU 4/5	FDP (median AIN branch + ulnar) and FCU (ulnar) are forearm muscles. They're less severely affected than hand intrinsics because some C8-T1 fibers to forearm muscles may be partially spared.
Triceps 5/5, Deltoid 5/5	These are C5-C7 muscles. Completely normal = upper trunk/posterior cord spared. The lesion is isolated to the lower trunk.
Medial arm/forearm/ulnar hand numbness	This is the combined territory of the medial brachial cutaneous, medial antebrachial cutaneous, and ulnar sensory nerves -- all from the medial cord/lower trunk.
NO Horner's	Horner's requires disruption of T1 sympathetic fibers at the ROOT level. Absence means the T1 root is intact -- the lesion is at the trunk (post-ganglionic).

WHY EACH NCS:

Study	Value	Why
Median sensory 45 uV (normal)	Median sensory to the index finger carries C6-C7 fibers. These come through the lateral cord / upper-middle trunk. Lower trunk injury doesn't affect this.	
Ulnar sensory 5 uV (ABNORMAL)	Ulnar sensory carries C8-T1 fibers through the lower trunk and medial cord. The lower trunk is damaged, so the sensory axons distal to the DRG have undergone Wallerian degeneration.	
Radial sensory 25 uV (normal)	Radial sensory carries primarily C6 fibers. Upper trunk is spared.	
MAC 4 uV (ABNORMAL)	THE KEY FINDING. MAC carries C8-T1 sensory from the medial cord. In C8-T1 radiculopathy, the MAC would be NORMAL (DRG is intact, peripheral axon survives). Here it's abnormal = POST-GANGLIONIC = the lesion is at the trunk or cord, not the root. This single test separates plexopathy from radiculopathy.	
Median motor (APB) 3.5 mV (LOW)	APB is C8-T1 via median nerve. The C8-T1 fibers destined for the median nerve travel through the lower trunk before reaching the medial cord. Lower trunk damage reduces the motor axons reaching APB.	
Ulnar motor (ADM) 2.5 mV (LOW)	ADM is C8-T1 via ulnar nerve. Same lower trunk pathway. Both median and ulnar motor are low = lesion proximal to where they split.	

WHY EACH EMG:

Muscle	Finding	Why
APB (Median, C8-T1)	2+ fibs	C8-T1 fibers through the lower trunk are damaged. APB loses its innervation and fibrillates.
FDI (Ulnar, C8-T1)	2+ fibs	Same C8-T1 pathway, different peripheral nerve. Both affected = proximal to the peripheral nerve level.
ADM (Ulnar, C8-T1)	2+ fibs	Another ulnar C8-T1 muscle. Confirms ulnar territory involvement.
EIP (Radial, C7-C8)	1+ fibs	THIS IS IMPORTANT. EIP is innervated by the radial nerve (posterior interosseous branch) at C7-C8. C8 fibers travel through the lower trunk. Even though the radial nerve comes from the posterior cord (not medial), some C8 fibers that will eventually reach the posterior cord pass through the lower trunk first. Mild involvement (1+ vs 2+) because EIP gets both C7 (spared) and C8 (affected).
FCU (Ulnar, C8)	1+ fibs	FCU is an ulnar forearm muscle. Less severe than hand intrinsics because it's a proximal muscle with potentially more surviving axons.
Biceps (Musculocutaneous, C5-C6)	Normal	Upper trunk muscle. Completely spared. Confirms the lesion is limited to the lower trunk.
Deltoid (Axillary, C5)	Normal	Same -- upper trunk muscle is fine.
C8-T1 Paraspinals	Normal (0 fibs)	Paraspinals are innervated by the posterior primary ramus at the root level. Normal paraspinals = the roots are intact = the lesion is POST-GANGLIONIC (at the trunk or below). This, combined with abnormal SNAPs, nails the localization to the lower trunk.

CASE 3: Flail Arm After Motorcycle Accident

Dx: Traumatic C5-T1 Root Avulsions

DIFFERENTIALS:

1. **Root Avulsions C5-T1** (correct)
2. **Cervical Spinal Cord Injury** -- ruled out because cord injury would cause bilateral symptoms or long tract signs (leg weakness, hyperreflexia, Babinski)
3. **Multifocal Motor Neuropathy** -- ruled out because MMN is distal-predominant and purely motor. This patient has sensory loss.
4. **Severe Diabetic Polyneuropathy** -- ruled out because polyneuropathy is symmetric and stocking-glove. This is unilateral and affects the entire arm.

WHY EACH PHYSICAL EXAM FINDING:

Finding	Why
0-1/5 throughout	ALL motor axons from C5-T1 are destroyed at the root level. No motor signals can reach any arm muscle.
Ptosis, miosis, anhidrosis (Horner's)	T1 root carries sympathetic fibers that exit the spinal cord and travel up the sympathetic chain to the superior cervical ganglion. Avulsing T1 disrupts these fibers BEFORE they leave. Ptosis = Mueller's muscle (sympathetic). Miosis = dilator pupillae (sympathetic). Anhidrosis = sweat glands on that side of face (sympathetic).
Winged scapula	Long thoracic nerve (C5-C7) innervates serratus anterior. All three root contributions are avulsed, so serratus anterior is completely denervated.
Sensation decreased C5-T1	The patient reports no sensation because the central axons from the DRG to the spinal cord are torn. The brain receives no sensory input from the arm.
All reflexes absent	Both the afferent (sensory) and efferent (motor) limbs of the reflex arcs are disrupted at the root level.

WHY EACH NCS -- THE PATHOGNOMONIC PATTERN:

Study	Value	Why
Median sensory 28 uV (NORMAL)	The DRG sits OUTSIDE the spinal canal in the neural foramen. Root avulsion tears the nerve BETWEEN the DRG and the cord. The peripheral sensory axon (DRG -> skin) is still connected to its cell body in the DRG. So it does NOT undergo Wallerian degeneration. The nerve conducts normally even though the patient feels nothing.	
Ulnar sensory 22 uV (NORMAL)	Same principle. The C8-T1 sensory neurons in the DRG are alive and well. Their peripheral axons are intact. Only the central projections to the cord are torn.	
Radial sensory 25 uV (NORMAL)	Same. All three major sensory nerves are preserved.	
All motor responses ABSENT	Motor neurons live in the ANTERIOR HORN of the spinal cord, not in a ganglion. The motor axons exit the cord and join the spinal nerve. Root avulsion tears these motor axons away from their cell bodies. Without their cell bodies, the axons undergo Wallerian degeneration. No motor axons = no CMAPs.	

The resident question: "How can SNAPs be normal if the patient can't feel anything?" Answer: Because the lesion is PROXIMAL to the DRG. The sensory cell body and its peripheral axon are intact (so the SNAP is normal), but the central axon connecting the DRG to the spinal cord is torn (so the brain receives no signal). It's like cutting a telephone wire between the switchboard and the central office -- the local phone still works, but nobody at headquarters can hear you.

WHY EACH EMG:

Muscle	Finding	Why
Every C5-T1 muscle	3+ fibs, no MUAPs	Every motor axon is dead. The muscle fibers are completely orphaned and fibrillating. "No voluntary MUAPs" means the patient cannot recruit a single motor unit -- there are none left.
Cervical Paraspinals C5-T1	3+ fibs	The posterior primary ramus branches off the spinal nerve almost immediately after it exits the foramen -- BEFORE the nerve enters the brachial plexus. If paraspinals are denervated, the lesion is at or proximal to the root level. In a plexopathy, paraspinals would be NORMAL because the posterior ramus branches off before the plexus. This finding CONFIRMS root-level pathology.

CASE 4: Sudden Groin Pain and Buckling Knee

Dx: Femoral Neuropathy from Retroperitoneal Hematoma

DIFFERENTIALS:

1. **Femoral Neuropathy from Hematoma** (correct)
2. **L3-L4 Radiculopathy** -- ruled out because adductors are normal (obturator nerve shares L2-L4 roots) AND paraspinals are normal
3. **Diabetic Amyotrophy** -- ruled out because patient is non-diabetic. Also amyotrophy is subacute/bilateral, not acute unilateral post-anticoagulation
4. **Lumbosacral Plexopathy** -- ruled out because plexopathy would affect BOTH femoral AND obturator distributions. Adductors are normal.
5. **Femoral Nerve at Inguinal Ligament** -- ruled out because **iliopsoas is weak**. Iliopsoas branches leave the femoral nerve BEFORE the inguinal ligament. If the lesion were at the ligament, iliopsoas would be spared.

WHY EACH PHYSICAL EXAM FINDING:

Finding	Why
Thigh larger on right	The hematoma in the iliacus muscle and retroperitoneum is expanding, causing visible swelling.
Hip held in flexion (psoas sign)	The iliopsoas muscle is being compressed by the hematoma. Stretching it (extending the hip) compresses the hematoma further = severe pain. Flexion relieves pressure. This is the classic "psoas sign."
Iliopsoas 2/5	Femoral nerve branches to iliopsoas arise WITHIN the iliac fossa, from the main trunk of the femoral nerve BEFORE it passes under the inguinal ligament. The hematoma is in the iliac fossa, compressing these branches.
Quadriceps 1/5	The femoral nerve's main motor branches to the quadriceps emerge AFTER the inguinal ligament. But the nerve is damaged proximal to the ligament, so the signal never reaches the quad.
Adductors 5/5 (NORMAL)	Adductors are innervated by the OBTURATOR nerve (L2-L4). The obturator nerve is a SEPARATE nerve from the femoral -- it comes from different roots of the lumbar plexus and takes a completely different path through the pelvis. A femoral nerve lesion CANNOT affect the obturator. If adductors were weak, you'd need to consider L3-L4 radiculopathy or plexopathy (which affect both).
Ankle muscles 5/5	Sciatic nerve territory (L4-S2) is completely uninvolved. The lesion is isolated to the femoral nerve.
Patellar reflex absent	The patellar reflex arc: femoral nerve (L2-L4) -> quadriceps -> patellar tendon. The efferent limb (femoral motor to quad) is destroyed.
Anterior thigh + medial leg numbness	Anterior thigh = femoral cutaneous branches. Medial leg = saphenous nerve (terminal sensory branch of femoral). Both are downstream of the femoral nerve.

WHY EACH NCS:

Study	Value	Why
Sural 16 uV (normal)	Sural = tibial nerve territory (S1-S2). Completely separate from the femoral nerve.	
Superficial Fibular 18 uV (normal)	Fibular nerve territory (L5-S1). Also separate.	
Saphenous ABSENT	The saphenous nerve is the terminal sensory branch of the femoral nerve. It carries L3-L4 sensory fibers. Since the femoral nerve is damaged proximal to the DRG (post-ganglionic at the nerve level), the saphenous sensory axons undergo Wallerian degeneration. An absent saphenous SNAP confirms POST-GANGLIONIC femoral nerve lesion. If this were L3-L4 radiculopathy (pre-ganglionic), the saphenous SNAP would be normal.	
Fibular/Tibial motor normal	Sciatic territory is fine.	
Femoral motor ABSENT on right	The femoral nerve's motor axons are completely blocked/destroyed by the hematoma. No signal reaches the quadriceps. Comparing to left side (6.5 mV) confirms unilateral pathology.	

WHY EACH EMG:

Muscle	Finding	Why
Iliopsoas 2+ fibs, severely reduced	THE KEY. Iliopsoas receives femoral nerve branches that originate IN the iliac fossa, PROXIMAL to the inguinal ligament. If iliopsoas is denervated, the lesion is in the retroperitoneum/iliac fossa -- NOT at the inguinal ligament. This is what separates a hematoma from a post-surgical compression.	
Vastus Lat/Med/Rectus Femoris 3+ fibs, no MUAPs	All quadriceps components are completely denervated. These all come from the femoral nerve. More severe than iliopsoas (3+ vs 2+, none vs severely reduced) suggests the main trunk is more affected than the early proximal branches.	
Adductor Longus (Obturator) NORMAL	This is the critical exclusion test. Adductor longus shares the same L2-L4 roots as the quadriceps but is innervated by the OBTURATOR nerve. If it's normal, the ROOTS are fine -- only the FEMORAL nerve is damaged. This rules out L2-L4 radiculopathy and lumbar plexopathy.	
Tibialis Anterior NORMAL	L4-L5 via deep peroneal nerve. Would be abnormal in L4 radiculopathy. Normal here = lesion is femoral nerve, not L4 root.	
Gluteus Medius NORMAL	L4-L5-S1 via superior gluteal. Would be abnormal in lumbar plexopathy. Normal = excludes plexopathy.	
Lumbar Paraspinals NORMAL	Posterior ramus branches off at the root. Normal paraspinals = not a radiculopathy.	

CASE 5: Foot Drop and Hamstring Weakness After Surgery

Dx: Sciatic Nerve Injury at the Hip (Post-THA)

DIFFERENTIALS:

1. **Sciatic Nerve Injury at Hip** (correct)
2. **Common Fibular Neuropathy at Fibular Head** -- ruled out because tibial-innervated muscles (gastrocnemius, tib posterior) and hamstrings are ALSO affected. Pure fibular palsy only hits peroneal-division muscles.
3. **L5-S1 Radiculopathy** -- ruled out because gluteal muscles (inferior gluteal = L5-S1-S2) and paraspinals are NORMAL
4. **Lumbosacral Plexopathy** -- ruled out because gluteal muscles (superior and inferior gluteal nerves arise from sacral plexus PROXIMAL to sciatic) are NORMAL
5. **Cauda Equina** -- ruled out because it would be bilateral with saddle anesthesia and bowel/bladder dysfunction

WHY EACH PHYSICAL EXAM FINDING:

Finding	Why
Calf + anterior compartment atrophy	6 weeks post-injury. Denervated muscles begin atrophying within 2-3 weeks. Both peroneal (anterior compartment) and tibial (calf) territories affected.
Equinus position (foot drop)	Tibialis anterior (dorsiflexion) is 0/5. Without dorsiflexion, gravity pulls the foot into plantarflexion.
Steppage gait	To clear the dropped foot during swing phase, the patient must lift the knee higher than normal.
Glut Med 5/5, Glut Max 5/5 (BOTH NORMAL)	Superior gluteal nerve (glut med) and inferior gluteal nerve (glut max) arise from the sacral plexus PROXIMAL to where the sciatic nerve forms. They exit the pelvis through the greater sciatic foramen but take separate paths from the sciatic nerve. A sciatic injury at the hip CANNOT affect these nerves. If they were weak, the lesion would have to be at the plexus or root level.
Hamstrings 2/5	The sciatic nerve gives motor branches to the hamstrings IN THE THIGH (before it reaches the knee). If hamstrings are weak, the lesion is proximal to the mid-thigh -- at the hip or above. This is what separates sciatic injury from fibular palsy at the knee.
TA 0/5, Gastroc 1/5, Peroneus 0/5, Tib Post 1/5	Both peroneal division (TA, peroneus) and tibial division (gastroc, tib post) muscles are affected. Peroneal division is MORE severely affected (0/5 vs 1/5) because the peroneal division has fewer fascicles and is more lateral in the sciatic trunk = more vulnerable to compression.
Saphenous sensation PRESERVED	Saphenous = femoral nerve territory. The femoral nerve is completely separate from the sciatic. Its preservation proves the lesion is isolated to the sciatic nerve.
Achilles absent, Patellar normal	Achilles reflex = tibial nerve (S1-S2). Destroyed because tibial division is damaged. Patellar reflex = femoral nerve (L2-L4). Normal because femoral nerve is intact.

WHY EACH NCS:

Study	Value	Why
Sural ABSENT	Sural = tibial division sensory (S1-S2). Post-ganglionic sciatic injury causes Wallerian degeneration of the sural sensory axons.	
Superficial Fibular ABSENT	Superficial fibular = peroneal division sensory (L5-S1). Same mechanism -- post-ganglionic axonal degeneration.	
Saphenous 8 uV (NORMAL)	Saphenous = femoral nerve. Completely separate from sciatic. Preserved = femoral nerve territory intact.	
Fibular motor (EDB) ABSENT	EDB = deep peroneal nerve. The motor axons from the peroneal division of the sciatic have completely degenerated.	
Fibular motor (TA) 0.4 mV	TA gets a recordable but severely reduced response. TA is a large proximal muscle -- sometimes a few surviving axons can still fire it weakly. Recording from TA (instead of the tiny EDB) is an alternative approach for severely affected nerves.	
Tibial motor (AH) 1.2 mV	AH = tibial nerve. Severely reduced but not absent. Tibial division is less severely affected than peroneal (1.2 mV vs 0) -- consistent with the peroneal division's greater vulnerability.	
Left side all normal	Side-to-side comparison proves the pathology is unilateral.	
H-reflex ABSENT on right	H-reflex = S1 reflex arc (tibial nerve afferent -> S1 synapse -> tibial nerve efferent). The sciatic injury disrupts both limbs of this arc on the right.	

WHY EACH EMG:

Muscle	Finding	Why
TA, EHL, Peroneus Longus	3+ fibs, NO MUAPs	Peroneal division muscles are completely denervated. 3+ fibrillations = severe acute-on-subacute denervation at 6 weeks. No MUAPs = no surviving motor units.
Gastroc, Tib Post	2+ fibs, severely reduced	Tibial division muscles are affected but LESS severely. 2+ fibs (not 3+) and "severely reduced" (not absent) recruitment. This asymmetry (peroneal worse than tibial) is classic for sciatic injury.
Biceps Femoris SHORT HEAD 2+ fibs	THE KEY MUSCLE. BFSH is the ONLY hamstring innervated by the PERONEAL DIVISION of the sciatic. The peroneal division gives off its branch to BFSH in the mid-thigh, BEFORE reaching the knee. If BFSH is denervated, the lesion is proximal to the mid-thigh (at the hip). In a common fibular palsy at the fibular head, BFSH would be COMPLETELY NORMAL because the branch to BFSH left the main nerve in the thigh. This single muscle distinguishes sciatic-at-hip from peroneal-at-knee.	
BF Long Head, Semimembranosus	1+ fibs	These are tibial-division hamstrings. Less severely affected (1+ vs 2+) consistent with the tibial-division-sparing pattern.
Gluteus Maximus NORMAL	Inferior gluteal nerve arises from the sacral plexus BEFORE the sciatic nerve forms. A sciatic injury at the hip CANNOT reach the inferior gluteal nerve. If glut max were weak, you'd need to look higher (plexus or root). This is THE localizing finding that rules out plexopathy.	
Gluteus Medius NORMAL	Superior gluteal nerve -- same principle as glut max. Branches off the plexus before the sciatic. Normal = lesion is below the plexus.	
TFL NORMAL	Also superior gluteal nerve. Additional confirmation.	
Vastus Lateralis NORMAL	Femoral nerve (L2-L4). Completely separate from sciatic.	
Lumbar Paraspinals NORMAL	Posterior ramus at the root level. Normal = NOT a radiculopathy.	